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AI Tool Prompt Engineering Report

In Partial Fulfillment of the GE 4120 - 21st Century IT Skills

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BSBM 2-C

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I

AI Tool Overview

Name of AI Tool: Google AI Overview

Purpose / Domain:

Google AI Overview is an artificial intelligence feature integrated into Google Search that generates summarized responses to user queries. It uses large language models (LLMs) to gather, synthesize, and present information from multiple sources in a clear and structured format. Its primary domain includes general knowledge retrieval, quick research, and everyday information seeking. It is designed to help users understand topics efficiently without needing to browse multiple websites.

Why you chose it:

I chose Google AI Overview because it is one of the most widely used and accessible AI-powered tools today. Almost everyone uses Google when looking for information, and the term “Google” has even become synonymous with the word “search.” This reflects how deeply embedded it is in everyday life and learning. Because of its familiarity and widespread use, it serves as a practical platform for analyzing how AI responds to different prompts. Additionally, using a tool that people already trust and rely on makes the findings of this activity more relevant and applicable in real-world contexts. It also allows for a clearer understanding of how prompt engineering can improve the quality of commonly encountered AI-generated outputs.

In addition, Google AI Overview works as a helpful research tool, similar to literature summarizers, because it takes information from many sources and presents it in a short and easy-to-understand form. This is useful for students like me who need to quickly understand key ideas. Using a tool that people already trust makes the results of this activity more practical and relevant. It also helps show how improving prompts can lead to clearer, more detailed, and more useful AI responses.

II

Original Prompt

Your initial prompt: Why is basketball important?

The AI-generated response explained that basketball is important because it provides a range of physical, mental, and social benefits. It highlighted improvements in cardiovascular health, strength, coordination, and endurance. The response also mentioned that basketball promotes teamwork, discipline, and social interaction. Additionally, it included benefits such as calorie burning, stress reduction, and confidence-building. Overall, the answer was correct but general in scope, as shown in Figure 1.

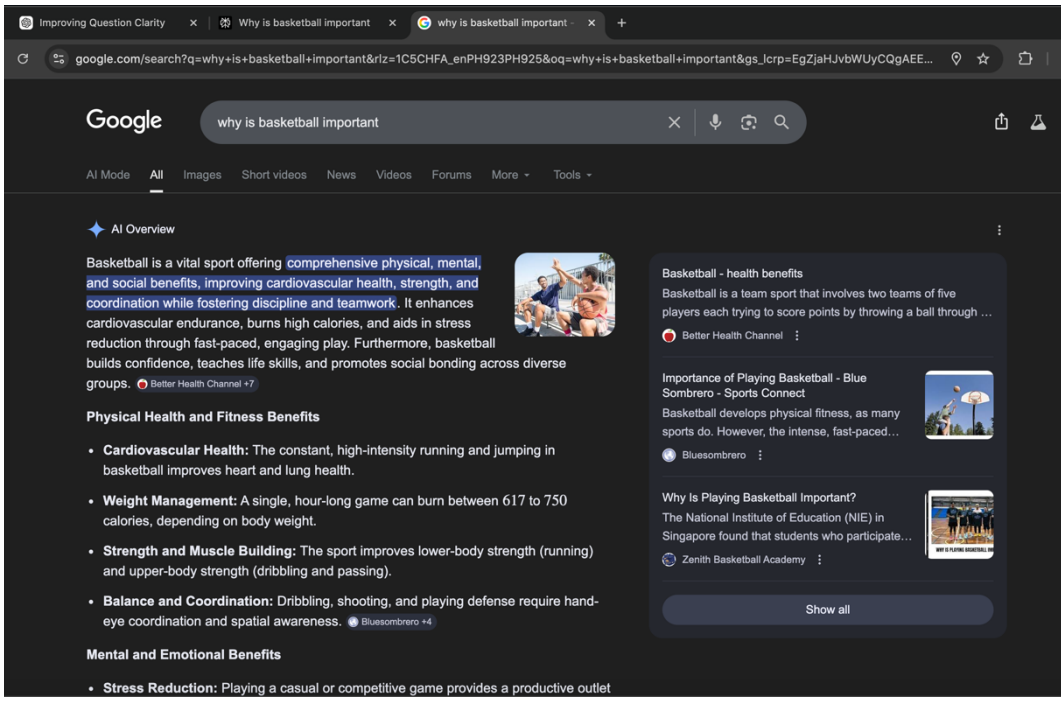

Figure 1. *Google AI Overview response to the original prompt "*Why is basketball important?*". The output presents a general summary highlighting physical, mental, and social benefits such as cardiovascular health, strength, coordination, teamwork, and stress reduction.*

III

AI-Assisted Prompt

Which LLM you used: ChatGPT

Improved prompt: Why is basketball important in people's lives?

The improved response provided a more comprehensive explanation of basketball's importance. It emphasized not only physical health benefits but also mental well-being, emotional development, and social connection. The output discussed how basketball fosters resilience, teamwork, and discipline while also building a sense of community and belonging. It also mentioned how the sport brings people together and shows how it connects people across cultures. Compared to the original, the response was more detailed, complete, and meaningful, as seen in Figure 2.

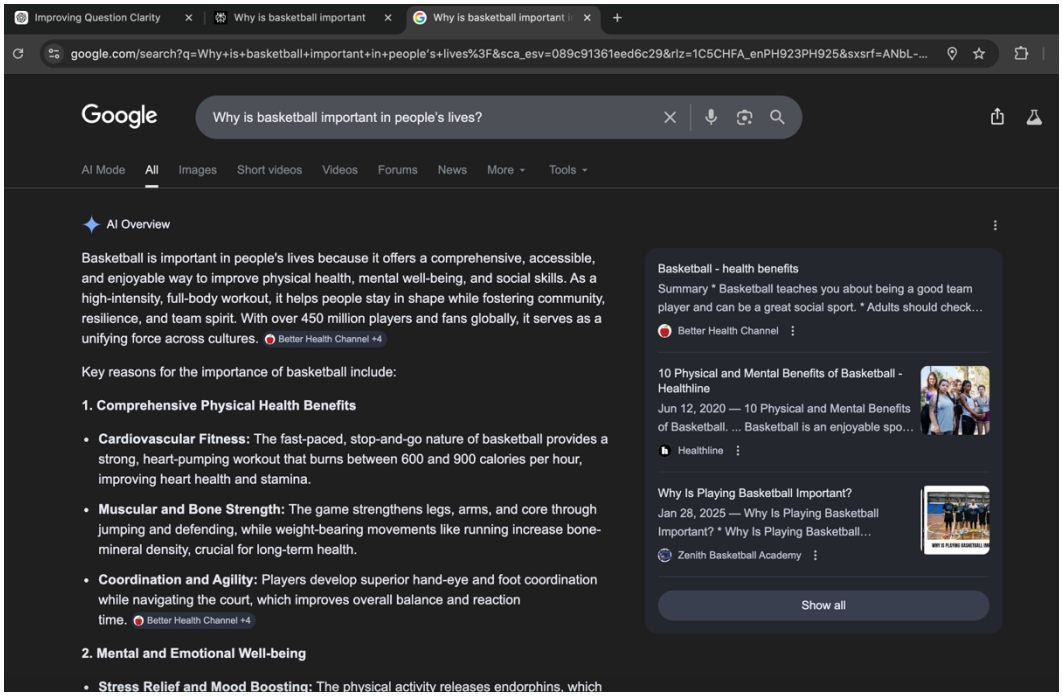

Figure 2. *Google AI Overview response to the improved prompt "*Why is basketball important in people's lives?*". The output provides a more detailed and contextualized explanation, emphasizing personal, emotional, social, and cultural significance, including community building, resilience, and overall well-being.*

IV
Comparison and Analysis

The outputs generated from the original and improved prompts demonstrate how prompt design significantly affects the quality of AI-generated responses. The original output (see Figure 1) was broad and focused mainly on general benefits such as physical health and teamwork. While accurate, it lacked deeper context and did not fully explore the broader significance of basketball.

On the other hand, the improved prompt, “Why is basketball important in people’s lives?”, produced a more detailed and useful answer. As seen in Figure 2, the response included not only physical benefits but also emotional, mental, and social aspects. It explained how basketball helps people grow, build relationships, and become more disciplined and resilient. This made the answer more meaningful and easier to relate to real-life situations.

The improved prompt made the response clearer because it gave more direction. By adding “in people’s lives,” the question became more specific. This helped the AI understand what kind of answer was needed. In terms of depth, the second response was better because it included more explanation and examples. In terms of relevance, it connected basketball to real-life experiences, making it more useful.

Furthermore, the use of ChatGPT to refine the prompt (as shown in Figure 3) demonstrates the importance of AI-assisted prompt engineering. The tool generated multiple variations of the question, helping identify a more effective and specific prompt. This process highlights how AI can support users in improving the clarity, depth, and relevance of their queries.

Generally, the improved prompt resulted in a clearer, more detailed, and more contextually relevant response. This activity shows that even small changes in wording can significantly enhance AI outputs. It also reinforces the importance of carefully designing prompts to maximize the effectiveness of AI tools.

From this activity, I learned that prompt engineering is very important when using AI. A clear and specific prompt can lead to better, more detailed, and more useful answers. It is important to think about how a question is asked, not just what is being asked. This shows that improving prompts can greatly improve the quality of AI-generated responses.

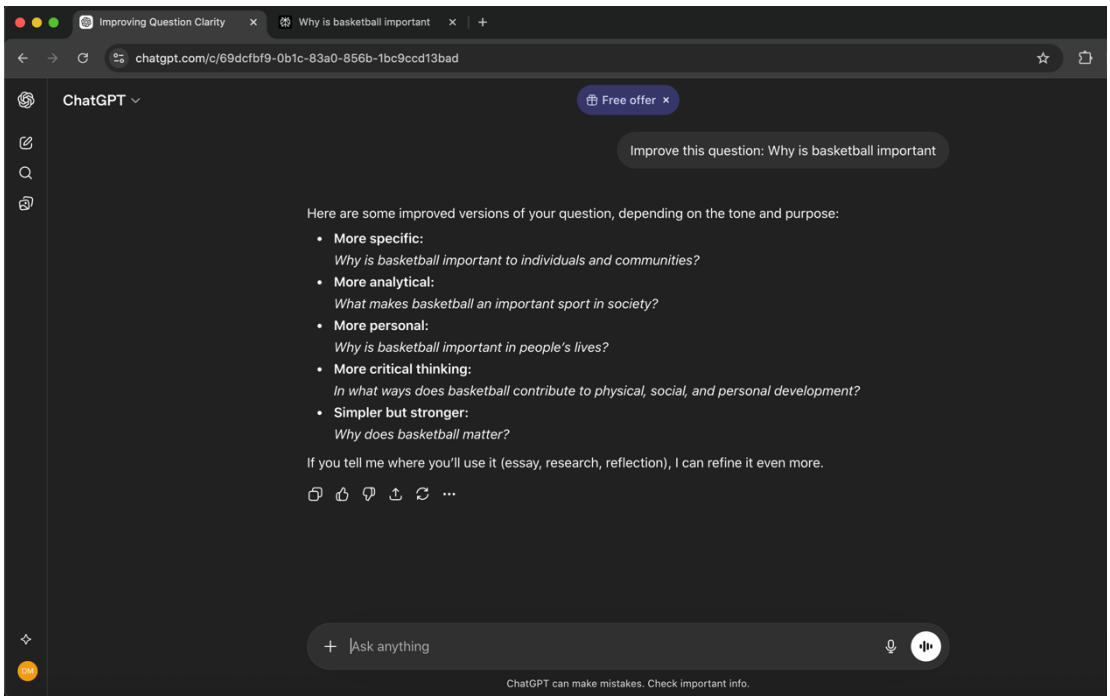

Figure 3. *Screenshot of ChatGPT generating improved versions of the original prompt "*Why is basketball important?*". The output shows multiple refined question formats categorized by tone and purpose, demonstrating how AI supports prompt engineering to enhance clarity, specificity, and depth.*